Daniel Kim

(505) 934-9722 | daniel360kim@gmail.com | daniel360kim.github.io

EDUCATION

Rice University **4.0/4.0**
Bachelor of Science in Electrical and Computer Engineering August 2024 – May 2028

- **Scholarships:** Max Roy Scholarship (1 recipient, full tuition, merit-based)
  Los Alamos Lab. Gold Scholarship (3 recipients, $20,000, merit, leadership, service-based)
- **Coursework:** Data Structures and Algorithms, Computer Vision, Machine Learning, OOP, Signals and Systems, Linear Algebra

SKILLS

Languages: C++, Python, Verilog, Dart, TypeScript/JavaScript, MATLAB/Simulink, Java, Swift, LaTeX
Frameworks: PyTorch, NumPy, Scikit-learn, Lightning, SciPy, Pandas, CUDA, OpenCV, OpenGL, ImGui, ROS
Other: Linux, Git, Docker, Google Cloud, Arduino, FPGA (Quartus, Vivado), Altium, KiCAD, Fusion360, I2C/SPI/UART, Keysight ADS

WORK AND RESEARCH EXPERIENCE

Rice Computational Imaging Lab **Houston, TX**
Undergraduate Research Assistant August 2025 – Present

- Architected Alpine, a modular PyTorch library to streamline the development of deep learning model development and training.
- Engineered a neural network pretraining method, boosting PSNR by 25-30 dB over meta-learned baselines, with 2x faster convergence.

NASA Kennedy Space Center **Cape Canaveral, FL**
Robotics Software Engineering Intern May 2025 – August 2025

- Developed a high-speed robotic vision system using ROS and OpenCV to track rocket trajectories in high-speed video.
- Pioneered a hybrid Neural-PID controller, decreasing tracking error by 94% and selected for deployment in Artemis moon missions.

Los Alamos National Laboratory **Los Alamos, NM**
Electrical Engineer R&D Intern June 2023 – August 2024

- Automated the validation pipeline for RF field stabilization, eliminating the need for $350k in legacy hardware resourced,
- Replaced manual PID tuning with robust self-tuning control algorithms for precise temperature control in particle accelerators.

Joint Science and Technology Institute West **Albuquerque, NM**
Computer Science Researcher June 2023

- Collaborated with DoE researchers to achieve a 92% runtime improvement in MPAS Ocean through tracer advection modeling.
- Engineered GPU/TPU-compatible heat transfer models by reformulating physics equations into efficient tensor algebra operations.

PUBLICATIONS

- Vyas, K., Kayabasi, A., **Kim, D.,** Saragadam, V., Veeraraghavan, A., & Balakrishnan, G. (2025). *The Surprising Effectiveness of Noise Pretraining for Implicit Neural Representations*. Submitted to IEEE/CVF CVPR 2026.
- **Kim, D**. (2024). *Engineered Underwater Vehicle for Ocean Litter Mapping.* New Mexico Journal of Science.
- **Kim, D.**, & Iturregui, A. (2022). *Developing a Control Algorithm and Simulation for Thrust Vector Controlled Rockets*. New Mexico Journal of Science.

PROJECTS

app.studybeats.co: Interactive web-based productivity suite (Dart, Google Cloud) November 2023 – August 2025

- Founded and launched StudyBeats, a real-time productivity platform with AI-driven study assistance and focus environments.
- Boosted system reliability for 1200+ users through optimized indexing, secure user authentication, and automated CI/CD workflows.

OceanAI: Underwater vehicle and computer vision for mapping ocean trash (C++, TypeScript, Python, KiCAD) June 2022 – May 2023

- Developed autonomous underwater vehicle to capture subsea footage and process it with YOLOv5-based object detection,
- Implemented deep learning and SLAM-based mapping to discover 174,000+ pieces of ocean trash, winning Regeneron ISEF 3rd place.

OrbitSim: Orbital simulator desktop app (C++, OpenGL, ImGUI). July 2023 – October 2023

- Simulated and visualized over 8000 active satellite trajectories and gravitational interactions using realistic physics models.
- Implemented a modular graphics pipeline with optimized memory usage, enabling smooth interactive 3D visualization and modeling.

NM Rocketry: Thrust vector-controlled model rocket (C++, PCB Design, Fusion360, Simulink) November 2021 – May 2022

- Engineered a real-time flight stabilization system, implementing PID with Kalman filter to actively control rocket trajectory.
- Developed PCB hardware and custom I2C/SPI drivers to achieve low-latency, high-reliability sensor data collection.